

Chapter 6

OpenShift Cluster Monitoring



Metrics collection and storage via Prometheus, an open-source monitoring system time series database.



Alerting/notification via Prometheus' Alertmanager, an open-source tool that handles alerts sent by Prometheus.



Metrics visualization via Grafana, the leading metrics visualization technology.



X sections in this chapter:

Section 6.1: Cloud Monitoring

Section 6.2: Guided Exercise

Section 6.3: Alerts and Notifications

Section 6.4: Guided Exercise

Section 6.5: Lab

Section 6.6: Summary

Clustering Monitoring



Chapter 6 Section 1



Objectives

After completing this section, you will be:

- Describe the architecture of
 - OpenShift Monitoring
- Query the
 - information in its dashboards.

OpenShift Observability

Logging

Aggregates all logs from pods and nodes.

- Centralized logging improves searching, visualization and reporting.

Monitoring

Collect metrics and generate alerts.

- Monitoring components and ensures they are always available and updated.

Network Observability

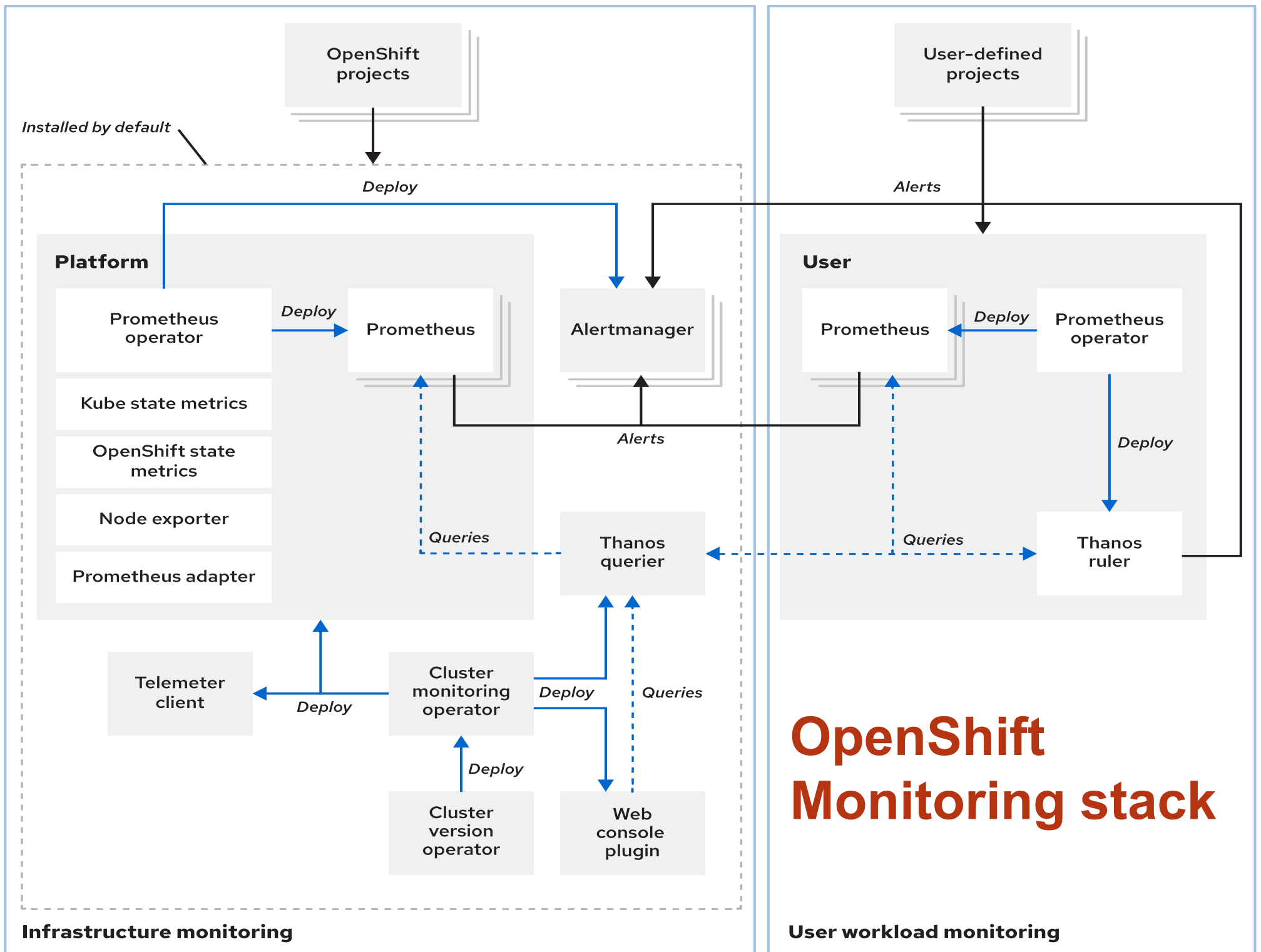
Monitors and analyzes network traffic

- Helps resolve network connectivity related issues.

Distributed Tracing

Monitoring, troubleshooting, and optimizing microservices

- tracking a single request as it moves through various services in distributed system.



Components in Default Monitoring Stack (1)

Cluster Monitoring Operator

- is the central component of the monitoring stack.
- controls the deployed monitoring components and ensures that they are always in sync with the latest version of the cluster monitoring operator.

Prometheus Operator

- The Prometheus operator deploys and configures both Prometheus and Alertmanager.
- The operator also manages the generation of configuration targets (service monitors and pod monitors).

Prometheus Server

- Runs the open-source Prometheus software.
- Collect telemetry from pods and nodes in cluster.

Prometheus Adapter

- Exposes cluster resources used in Horizontal Pod Autoscaling (HPA).

Components in Default Monitoring Stack (2)

Prometheus Alertmanager

- Handles configured alerts. Send out alerts as notifications.

Kube state metrics

- Exports and format Kubernetes objects to metrics parseable by Prometheus.

OpenShift state metrics

- Expands upon kube-state-metris by adds monitoring for OpenShift-specific resources.

Node exporter

- Exports low-level metrics for compute nodes.

Thanos Querier

- is a single, multitenant interface that enables aggregating and deduplicating cluster and user workload metrics.

Components in Default Monitoring Stack (3)

Thanos Ruler

- Evaluates rules for Prometheus components

Telemeter Client

- sends a data portion from Prometheus instances to Red Hat for remote health monitoring.

The monitoring web console

- Provides the visualization using web user interface.
- Provides the Observe section to access and manage monitoring features:
 - can access monitoring dashboards, metrics, alerts, and metrics targets.

Components that runs in user-defined project

Prometheus Operator



Prometheus



Thanos Ruler



Alertmanager



Prometheus

- Open-source project for system monitoring and alerting.
- Used by both Kubernetes and Red Hat OpenShift Platform.
- Gathers and stores telemetry from cluster in time-series data.
 - A timestamp.
 - A numeric value (integer, float or Boolean).
 - A set of labels in key/value pairs form.
- Labels can be used:
 - by Thanos Querier,
 - for filtering/grouping/aggregation.
- ❖ Example: `machine_cpu_cores` metric
 - Sequence of usage on number of CPU cores for each compute nodes.

OpenShift Monitoring Web Console

OpenShift Container Platform web console provides the Observe section, with the following subsections:

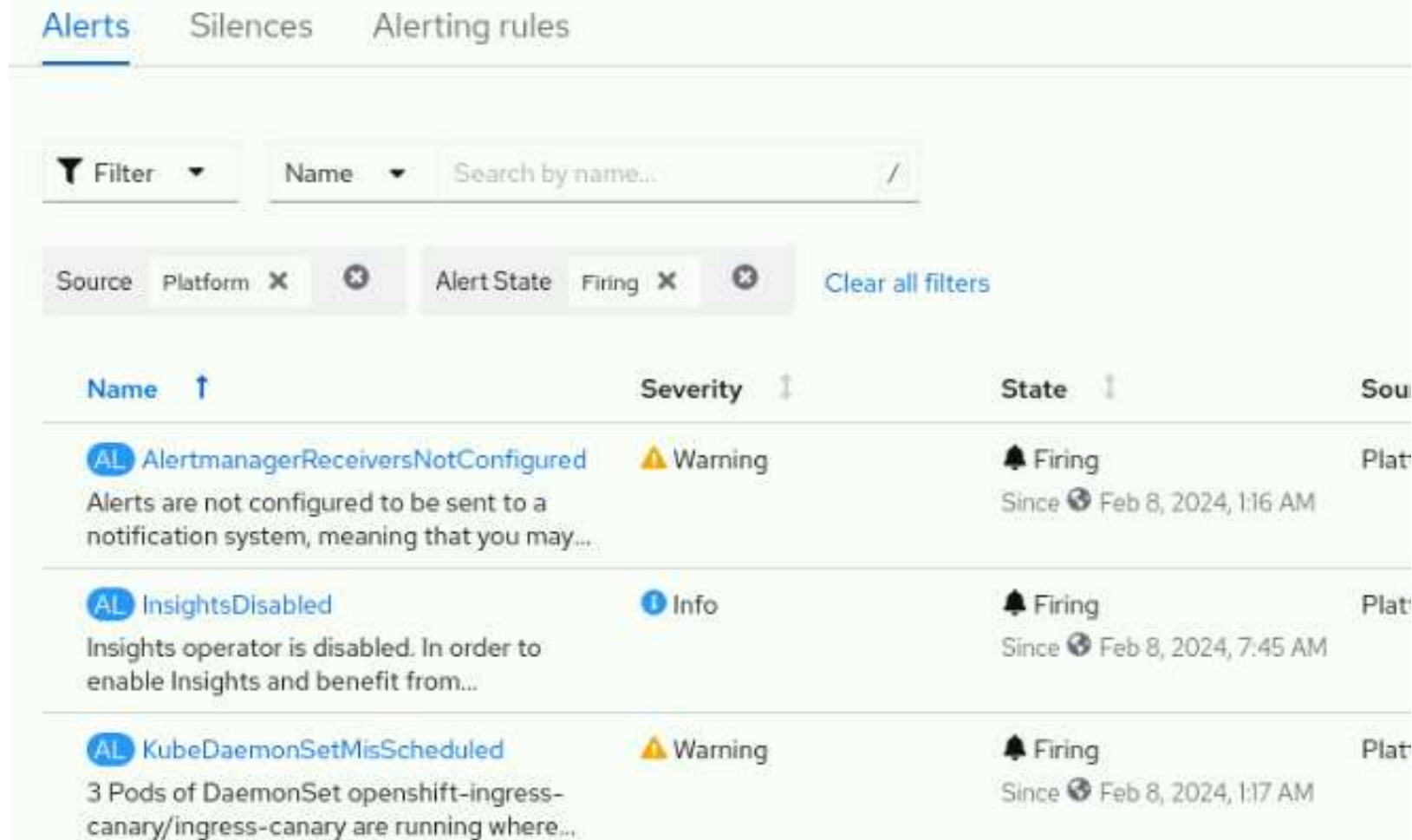
- Alerting
- Metrics
- Dashboards
- Targets

OpenShift Monitoring Web Console

OpenShift Container Platform web console provides the Observe section, with the following subsections:

✓ Alerting

- Metrics
- Dashboards
- Targets



Alerts Silences Alerting rules			
Filter Name Search by name... /			
Source Platform X Alert State Firing X Clear all filters			
Name ↑	Severity ↑	State ↑	Source
AL AlertmanagerReceiversNotConfigured Alerts are not configured to be sent to a notification system, meaning that you may...	Warning	Firing Since Feb 8, 2024, 1:16 AM	Platform
AL InsightsDisabled Insights operator is disabled. In order to enable Insights and benefit from...	Info	Firing Since Feb 8, 2024, 7:45 AM	Platform
AL KubeDaemonSetMisScheduled 3 Pods of DaemonSet openshift-ingress-canary/ingress-canary are running where...	Warning	Firing Since Feb 8, 2024, 1:17 AM	Platform

OpenShift Monitoring Web Console

OpenShift Container Platform web console provides the Observe section, with the following subsections:

- Alerting
- ✓ **Metrics**
- Dashboards
- Targets



Buttons: Add query, Run q

Query: `instance:node_cpu_utilisation:rate1m`

Unselect all

	Name	container	endpoint	instance	job	namespace	pod
☒	instance:node_cpu_utilisation:rate1m	kube-rbac-proxy	https	master01	node-exporter	openshift-monitoring	node-exporter

PromQL - Prometheus query language

- Filter metric to include certain labels

```
instance:node_cpu_utilisation:rate1m{instance="worker02"}  
kube_persistentvolumeclaim_status_phase{phase="Pending"} == 1
```

- Sort metric ascending or descending.

```
sort_by_label(v instant-vector, label string, ...):  
sort_by_label_desc(v instant-vector, label string, ...):
```

- Compute aggregations on time-series data.

- **sum()**

- Adds the value of all sample entries at a given time.

- **rate()**

- Computes the per-second average of a time series for a given time range.

- **count()**

- Counts the number of sample entries at a given time.

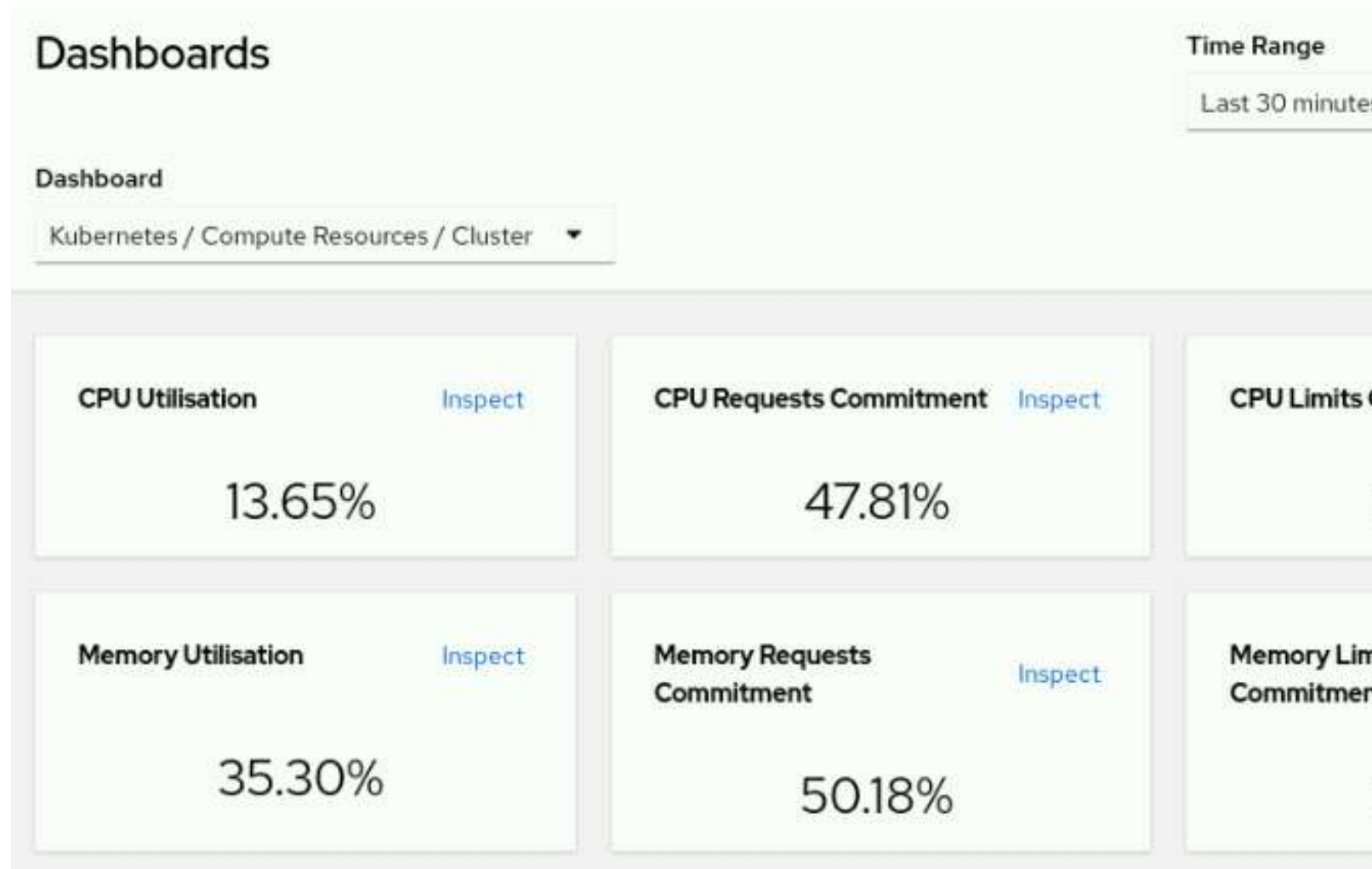
- **max()**

- Selects the maximum value out of the sample entries.

OpenShift Monitoring Web Console

OpenShift Container Platform web console provides the Observe section, with the following subsections:

- Alerting
- Metrics
- ✓ **Dashboards**
- Targets



OpenShift Monitoring Web Console

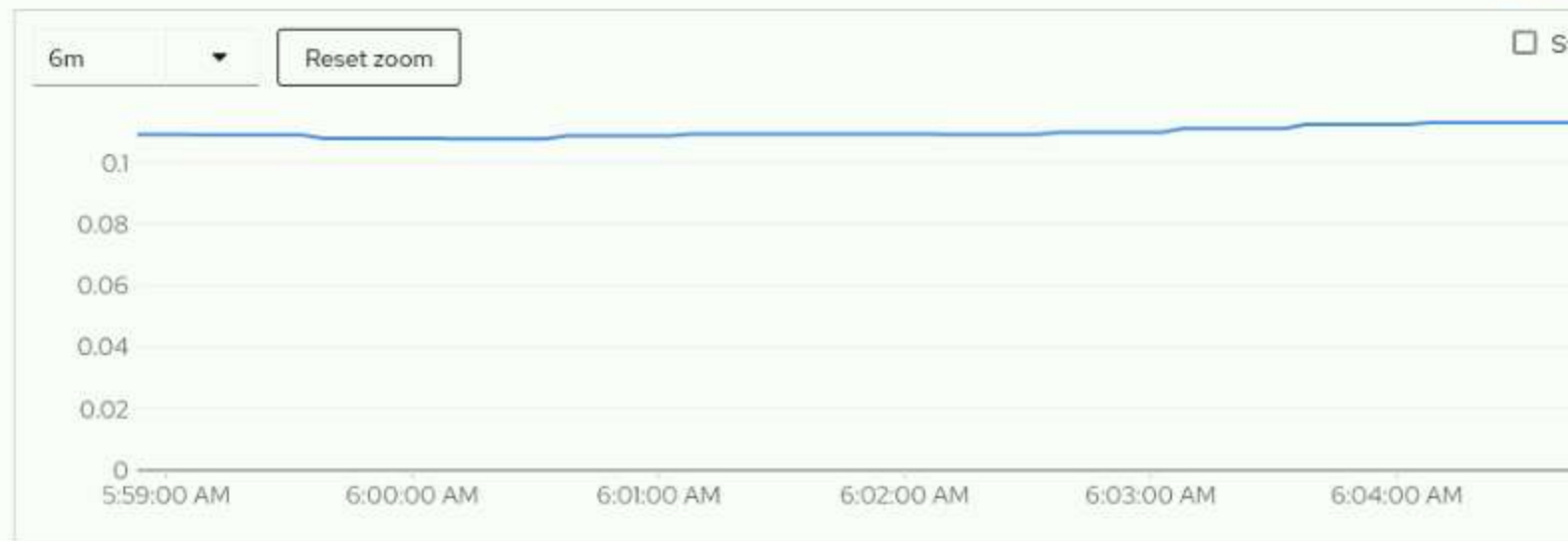
OpenShift Container Platform web console provides the Observe section, with the following subsections:

- Alerting
- Metrics
- ✓ Dashboards
- ✓ **Targets**

Metrics

Refresh off ▼

Action



Add query

Run q



cluster:node_cpu:ratio_rate5m{cluster=""}



Unselect all

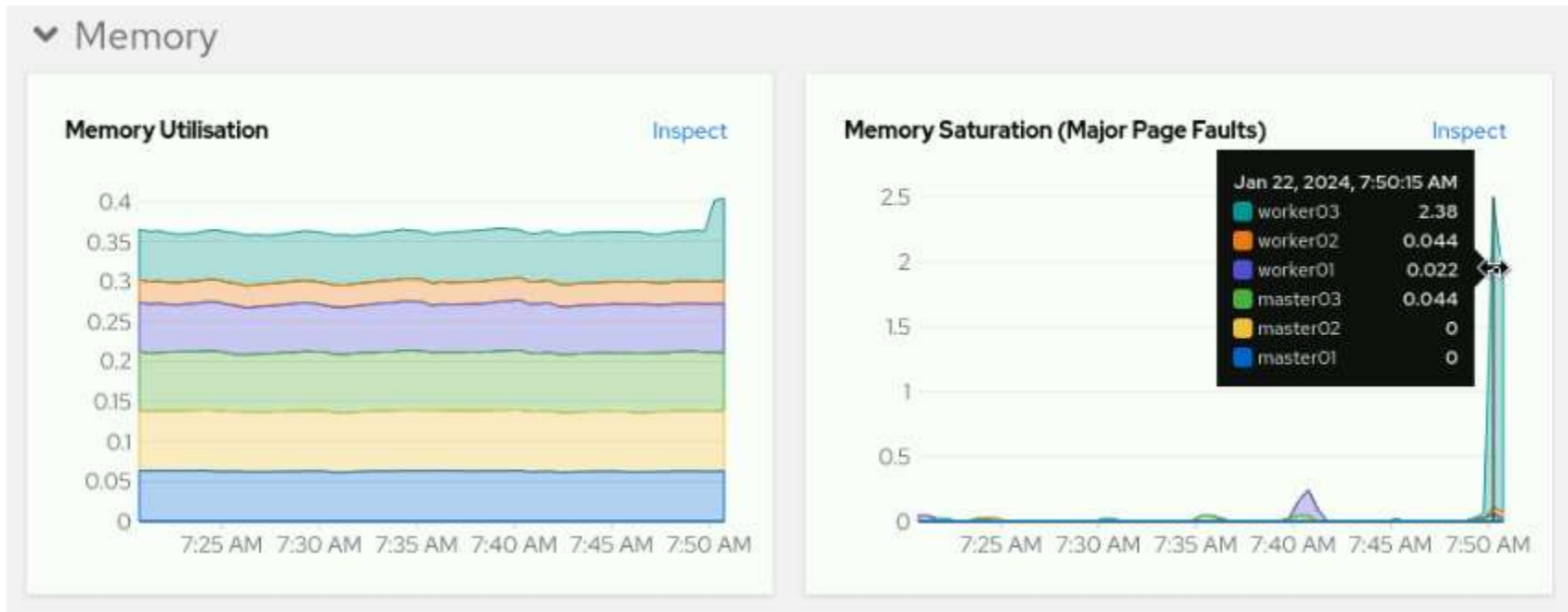
Name

prometheus

Value

USE Method / Cluster

- Utilization Saturation and Errors (USE).
- Identify cluster node for
 - **Overutilized** : excessive use of specific resource.
 - **Oversaturated**: specific node filled beyond its capacity; leaving no room for growth.
 - or experiencing many errors.



Section summary



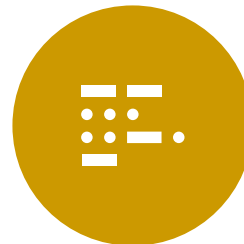
Reviewed components in
OpenShift Monitoring stack



Understand Prometheus



Explore OpenShift Monitoring
Web Console



Review PromQL

Section 6.2: Guided Exercise Cluster Monitoring

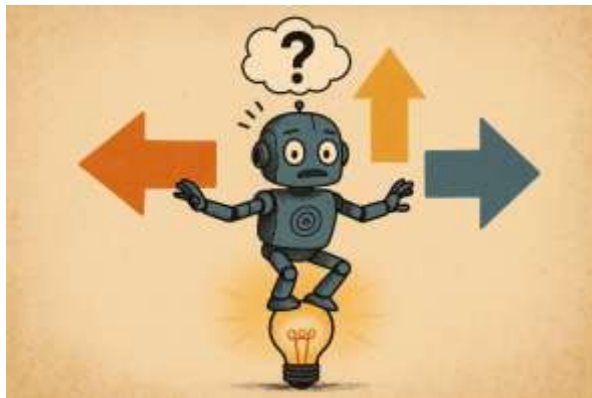
You should be able to:

- Use the OpenShift monitoring default stack to identify a deployment that consumes excessive CPU and memory.
- Use the OpenShift monitoring default stack to find the cause of an availability issue.

Estimated completion time: 20 - 30 minutes



**Continue /
Guided Exercise /
Break?**



Alerts and Notifications



Chapter 6 Section 2



Objectives

After completing this section, you will be:

- View alerting rules.
- Configure and silence alerts.

OpenShift Alerts

Alerting rules

- A set of conditions that outline a particular state within a cluster.
- When conditions met, trigger alerts.
- An alerting rule can be assigned a severity that defines how the alerts are routed.

Alerts

- Triggered when conditions that are defined in an alerting rule are true.
- Execute notification to receivers

Alert receivers

- Email, send pager notifications, or forward them to another system by using a webhook.

Silences

- A silence can be applied to alert to prevent sending notifications when the conditions for an alert are true.
- Avoid receiving notification while work on resolving underlying issue.

Viewing Alerting Rules

From web console, click **Observe** → **Alerting** → **Alerting rules** tab.

Alerting

Alerts Silences Alerting rules

Filter ▼

Name ▼ PersistentVolume

Source Platform X + Name PersistentVolume X Clear all filters

Name ↑	Severity ↓	Alert state ↓	Source ↓
AR KubePersistentVolumeFillingUp	Critical	-	Platform
AR KubePersistentVolumeFillingUp	Warning	-	Platform
AR PersistentVolumeUsageCritical	Critical	1 Firing	Platform
AR PersistentVolumeUsageNearFull	Warning	1 Silenced	Platform

Alerting rule details page (1)

- Description, message, labels, and PromQL expression.

[Alerting rules](#) > Alerting rule details

AR PersistentVolumeUsageNearFull  Warning

Alerting rule details

Name

PersistentVolumeUsageNearFull

Source

Platform

Severity

 Warning

For

5s

Description

PVC `{{ $labels.persistentvolumeclaim }}` utilization has crossed 75%. Free up some space or expand the PVC.

Expression

```
(kubelet_volume_stats_used_bytes * on (namespace, persistentvolumeclaim)
group_left (storageclass, provisioner) (kube_persistentvolumeclaim_info
* on (storageclass) group_left (provisioner)
kube_storageclass_info{provisioner=~"(. *rbd.csi.ceph.com)|
(. *cephfs.csi.ceph.com)"})) / (kubelet_volume_stats_capacity_bytes * on
(namespace, persistentvolumeclaim) group_left (storageclass,
provisioner) (kube_persistentvolumeclaim_info * on (storageclass)
group_left (provisioner) kube_storageclass_info{provisioner=~"
(. *rbd.csi.ceph.com)|(. *cephfs.csi.ceph.com)"})) > 0.75
```

Message

PVC `{{ $labels.persistentvolumeclaim }}` is nearing full. Data deletion or PVC expansion is required.

Labels

prometheus=openshift-monitoring/k8s

severity=warning

Alerting rule details page (2)

- Scrolls down to view graph

Alerting rules > Alerting rule details

AR PersistentVolumeUsageNearFull Warning



Active alerts

30m



Reset zoom

View in Metrics

0.8
0.6
0.4
0.2
0

7:30 PM

7:35 PM

7:40 PM

7:45 PM

7:50 PM

7:55 PM

Description

Active since

State

Value

PVC mysql utilization has crossed 75%. Free up some space or expand the PVC.

Jan 31, 2024, 6:08 PM

Firing

9.8828125e-01

Viewing Firing Alerts

From web console, click **Observe** → **Alerting** → **Alerts** tab.

Alerts Silences Alerting rules

Filter

Name

Search by name...

Source

Platform

Alert State

Firing

Clear all filters

Name

Severity

State

Source

AL PersistentVolumeUsageCritical

Critical

Firing

Platform

PVC mysql utilization has crossed 85%.
Free up some space or expand the PVC immediately.

Since Jan 26, 2024, 5:09 PM

AL PersistentVolumeUsageNearFull

Warning

Firing

Platform

PVC mysql utilization has crossed 75%.
Free up some space or expand the PVC immediately.

Since Jan 26, 2024, 5:09 PM

AL Watchdog

None

Firing

Platform

Alert details (1)

- Graph displays time-series data when alert was triggered/fired.

[Alerts](#) > Alert details

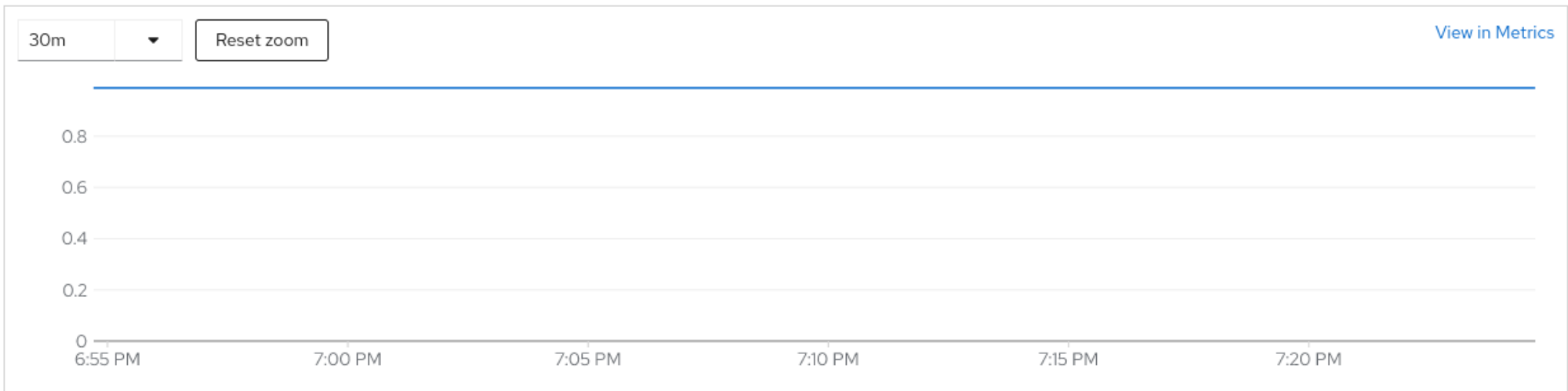
AL PersistentVolumeUsageNearFull Warning

Silence alert

PVC mysql utilization has crossed 75%. Free up some space or expand the PVC.

Alert details

[Hide graph](#)



Alert details (2)

- Scrolls down to view alert details, description, labels and etc.

Alerts > Alert details

AL PersistentVolumeUsageNearFull ⚠ Warning

Silence alert

PVC mysql utilization has crossed 75%. Free up some space or expand the PVC.

Alert details

Name

PersistentVolumeUsageNearFull

Source

Platform

Severity

⚠ Warning

State

🔔 Firing

Since 🌐 Jan 26, 2024, 5:09 PM

Description

PVC mysql utilization has crossed 75%. Free up some space or expand the PVC.

Message

PVC mysql is nearing full. Data deletion or PVC expansion is required.

Labels

alertname=PersistentVolumeUsageNearFull endpoint=https-metrics instance=192.168.50.13:10250 job=kubelet metrics_path=/metrics namespace=monitoring-alerts node=worker
persistentvolumeclaim=mysql provisioner=openshift-storage.cephfs.csi.ceph.com service=kubelet severity=warning storageclass=ocs-external-storagecluster-cephfs

Alerting rule

AR PersistentVolumeUsageNearFull



Silence Alerts

- Silence alert for period of time
- Stop new notifications sent to receiver.
- Case use: while troubleshooting a problem or maintenance window.

Alerts Silences Alerting rules

Filter Name Search by name... /

Source Platform X Alert State Firing X Clear all filters

Name ↑	Severity ↑	State ↑	Source
AL PersistentVolumeUsageCritical PVC mysql utilization has crossed 85%. Free up some space or expand the PVC immediately.	Critical	Firing Since Jan 26, 2024, 5:09 PM	Platform
AL PersistentVolumeUsageNearFull PVC mysql utilization has crossed 75%. Free up some space or expand the PVC immediately.	Warning	Firing Since Jan 26, 2024, 5:09 PM	Platform
AL Watchdog This is an alert meant to ensure that the entire alerting	None	Firing Since Jan 26, 2024, 2:18 PM	Platform

Silence alert

View alerting rule

Silence Alerts

- Silence alert for period of time
- Start immediately or set specific start time.
- Case use: while troubleshooting a problem or maintenance window.

Silence alert

Silences temporarily mute alerts based on a set of label selectors that you define. Notifications will not be sent for alerts that match all the listed values or regular expressions.

Duration

Silence alert from...

Now

For...

30m

Until...

30m from now

☒ Start immediately

Alert labels

Alerts with labels that match these selectors will be silenced instead of firing. Label values can be matched exactly or with a [regular expression](#)

Label name

alertname

Label value

PersistentVolumeUsageNearFull

☐ RegEx ☐ Negative matcher



Use RegEx

- **Silence all alerts from a specific service:**
 - `service =~ web-prod|api-prod`
- **Silence all alerts except "Watchdog":**
 - `alertname !~ Watchdog.`
- **Silence alerts from all instances in a Europe region:**
 - `cluster =~ europe-.*`
- **Silence alerts with any value for a specific label:**
 - `environment =~ .+`
 - (matches any non-empty string for the 'severity' label, effectively matching all alerts that have that label present). Note that using `.*` for matching *everything* might not work in all Alertmanager versions, so `.+` is safer for non-empty values.

Viewing Alert Silences

- Click **Observe** → **Alerting** → **Silences** tab.

[Alerts](#) [Silences](#) [Alerting rules](#)

Filter

Search by name... /

Create silence

Expire 0 silences

Silence State

Active X

Pending X

X

[Clear all filters](#)

☐	Name ↑	Firing alerts ↑	State ↑	Creator ↑
☐	SL PersistentVolumeUsageNearFull alertname=PersistentVolumeUsageNearFull endpoint=https-metrics instance=192.168.50.13:10250 job=kubelet metrics_path=/metrics severity=warning namespace=monitoring-alerts node=worker01	⚠ 1	✔ Active Ends 🌐 Feb 2, 2024, 5:19 PM	admin ⋮

Silence details page

Silences > Silence details

SL PersistentVolumeUsageNearFull

Actions ▾

Silence details

Name

PersistentVolumeUsageNearFull

Starts at

🌐 Feb 2, 2024, 4:49 PM

Matchers

alertname=PersistentVolumeUsageNearFull

endpoint=https-metrics

severity=warning

instance=192.168.50.13:10250

job=kubelet

metrics_path=/metrics

service=kubelet

namespace=monitoring-alerts

node=worker01

persistentvolumeclaim=mysql

provisioner=openshift-storage.cephfs.csi.ceph.com

storageclass=ocs-external-storagecluster-cephfs

Ends at

🌐 Feb 2, 2024, 5:19 PM

Created by

admin

Comment

Silenced during troubleshooting

State

✔ Active

Firing alerts

⚠ 1

Viewing Silenced Alerts

- Clear the Alert State filter by clicking **x** to display all alerts.

Alerts Silences Alerting rules

Filter Name Search by name...

Source Platform **x** **x** Alert State Firing **x** **x** Clear all filters

Name ↑	Severity ↑	State ↑	Source ↑
AL PersistentVolumeUsageCritical PVC mysql utilization has crossed 85%. Free up some space or expand the PVC immediately.	! Critical	🔔 Firing Since 🌐 Jan 26, 2024, 5:09 PM	Platform ⋮
AL Watchdog	i None	🔔 Firing	Platform ⋮

Viewing Silenced Alerts

- Clear the Alert State filter by clicking **x** to display all alerts.

Alerts Silences Alerting rules

Filter Name Search by name...

Source Platform X Alert State Firing X Clear all filters

Name ↑	Severity ↑	State ↑	Source ↑
AL PersistentVolumeUsageCritical PVC mysql utilization has crossed 85%. Free up some space or expand the PVC immediately.	Critical	Firing Since Jan 26, 2024, 5:09 PM	Platform
AL Watchdog	None	Firing	Platform

Viewing Silenced Alerts

- Clear the Alert State filter by clicking **x** to display all alerts.

Alerts Silences Alerting rules

Filter				
Name Search by name...				
Source Platform X Clear all filters				
Name ↑	Severity ↑	State ↑	Source ↑	
AL PersistentVolumeUsageCritical PVC mysql utilization has crossed 85%. Free up some space or expand the PVC immediately.	 Critical	 Firing Since  Jan 26, 2024, 5:09 PM	Platform	
AL PersistentVolumeUsageNearFull PVC mysql utilization has crossed 75%. Free up some space or expand the PVC immediately.	 Warning	 Silenced Ends  Jan 26, 2024, 8:16 PM	Platform	
AL Watchdog	 None	 Firing	Platform	

Sending Alert Notifications

- Not configured by default.
- Configure alert to send to following receiver types:
 - **Email**
 - sent to a specified address or distribution list.
 - **Microsoft Teams**
 - Sends alerts to a Microsoft Teams channel.
 - **PagerDuty**
 - Requires PagerDuty integration key, for immediate attention.
 - **Slack**
 - Sends notifications directly to a Slack channel.
 - **Webhook**
 - Allows you to send alerts to a custom webhook endpoint, which can then be routed to other systems for processing.
- OpenShift also provides a debug receiver, which can be helpful for development and testing purposes to see the alert details in a deployment.

Alertmanager

- Alerting systems uses Alertmanager (component of Prometheus server)
 - Send notification to receiver.
- Configure via
 - web console
 - oc command line
 - Edit alertmanager-main **secret** configuration stores in openshift-monitoring namespace.

Configuring Alertmanager with Web Console

- Click **Administration** → **Cluster Settings** → **Configuration** tab → **Alertmanager**.

Cluster Settings

Details ClusterOperators Configuration

Filter by name or description... /

Edit the following resources to manage the configuration of your cluster.

Configuration resource

Description

Alertmanager

Configure grouping and routing of alerts

Authentication
config.openshift.io

Authentication specifies cluster-wide settings for authentication (like webhooks and token authenticators). The canonical name of an instance is `config.openshift.io`.
Compatibility level 1: Stable within a major release for a minimum of 12 months.

Create Receiver
Edit configuration YAML

Authentication

Authentication provides information to configure an operator to manage

Alertmanager configuration page

- Lists current alert routing parameters:
 - Group by namespace / node
 - Group wait – waiting time before send first notification.
 - Group interval – iminimum wait time before send new alert.
 - Repeat interval – minimum wait time in between repeating alert (same condition)

[Configuration](#) > Alertmanager details

Alertmanager

[Details](#) [YAML](#)

Alert routing

Edit

Group by
namespace

Group interval
5m

Group wait
30s

Repeat interval
12h

Alertmanager configuration page

- Scrolls down to view configured alert receivers:
 - Configure additional receiver.
 - Edit or delete receiver.

[Configuration](#) > [Alertmanager details](#)

Alertmanager



Receivers

Filter Receivers by name...

Create Receiver

Name

Integration type

Routing labels

Critical

 [Configure](#)

Default

 [Configure](#)

All (default receiver)

null

Edit Receiver

Watchdog

Delete Receiver

View and edit Alertmanager using YAML

- YAML can be part of GitOps.

[Configuration](#) > [Alertmanager details](#)

Alertmanager

Details [YAML](#)

Update this YAML to configure Routes, Receivers, Groupings and other Alertmanager settings.

Alt + F1

[Accessibility help](#)

☒ [Show tooltips](#)

```
1 global:
2   resolve_timeout: 5m
3 inhibit_rules:
4   - equal:
```

✓ alertmanager-main has been updated to version 123456789

Save

Cancel

 Download

Configuring Alertmanager with CLI

- Alertmanager configuration is saved in the alertmanager-main **secret** in the openshift-monitoring namespace.

- Verify secret in openshift-monitoring namespace:

```
$ oc get secret/alertmanager-main -n openshift-monitoring
```

NAME	TYPE	DATA	AGE
alertmanager-main	Opaque	1	7d

- Extract the secret and save it in yaml format file”

```
$ oc extract secret/alertmanager-main -n openshift-monitoring --to ./ --confirm  
alertmanager.yaml
```

- Modify the yaml file accordingly.

Sample of modified alertmanager.yaml (1)

```
global:
  resolve_timeout: 5m
  smtp_from: alerts@ocp4.example.com 1
  smtp_smarthost: '192.168.50.254:25' 2
  smtp_hello: localhost 3
  smtp_auth_username: smtp_training 4
  smtp_auth_password: Red_H4T@! 5
  smtp_require_tls: false 6
  ...output omitted...
inhibit_rules:
  ...output omitted...
receivers:
  ...output omitted...
  - name: email 7
    email_configs: 8
      - to: ocp-admins@example.com 9
```

Sample of modified alertmanager.yaml (2)

```
global:
  resolve_timeout: 5m
  smtp_from: alerts@ocp4.example.com
  smtp_smarthost: '192.168.50.254:25'
  smtp_hello: localhost
  smtp_auth_username: smtp_training
  smtp_auth_password: Red_H4T@!
  smtp_require_tls: false
...output omitted...
inhibit_rules:
  ...output omitted...
receivers:
  ...output omitted...
  - name: email
    email_configs:
      - to: ocp-admins@exam
```



```
route:
  group_by:
    - namespace
  group_interval: 2m
  group_wait: 30s
  receiver: Default
  repeat_interval: 1m
  routes:
    ...output omitted...
    - receiver: email
      match:
        alertname: PersistentVolumeUsageNearFull
        continue: true
    ...output omitted...
```

Update Alertmanager

```
$ oc set data secret/alertmanager-main -n openshift-monitoring \
  --from-file alertmanager.yaml
```

```
secret/alertmanager-main data updated
```

- Ensure no mistakes in yaml and view log generate “Completed loading of configuration file” message.

```
$ oc logs -f statefulset.apps/alertmanager-main \
  -c alertmanager -n openshift-monitoring
```

```
Found 2 pods, using pod/alertmanager-main-0
```

```
...output omitted...
```

```
ts=2024-01-31T01:02:03.064Z caller=coordinator.go:113 level=info component=configuration
  msg="Loading configuration file" file=/etc/alertmanager/config_out/alertmanager.env.yaml
```

```
ts=2024-01-31T01:02:03.128Z caller=coordinator.go:126 level=info component=configuration
  msg="Completed loading of configuration file"
  file=/etc/alertmanager/config_out/alertmanager.env.yaml
```


Sample of misconfiguration

...output omitted...

```
ts=2024-01-31T02:03:04.256Z caller=coordinator.go:113 level=info component=configuration  
msg="Loading configuration file" file=/etc/alertmanager/config_out/alertmanager.env.yaml
```

```
ts=2024-01-31T02:03:04.512Z caller=coordinator.go:118 level=error component=configuration  
msg="Loading configuration file failed" file=/etc/alertmanager/config_out/alertmanager.env.yaml  
err="no global SMTP smarthost set"
```

- If you see configuration errors in the logs, then modify the alertmanager.yaml file and reapply your changes to the alertmanager-main secret in the openshift-monitoring namespace.

Section summary



Understand needs for alerts



Use web console to view alert and alert rules



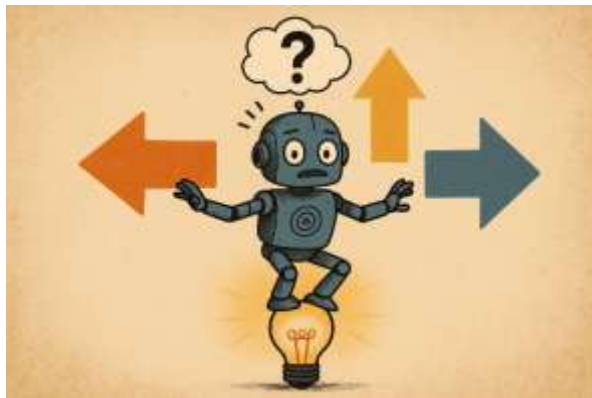
Silence alerts



Configure Alert Receivers

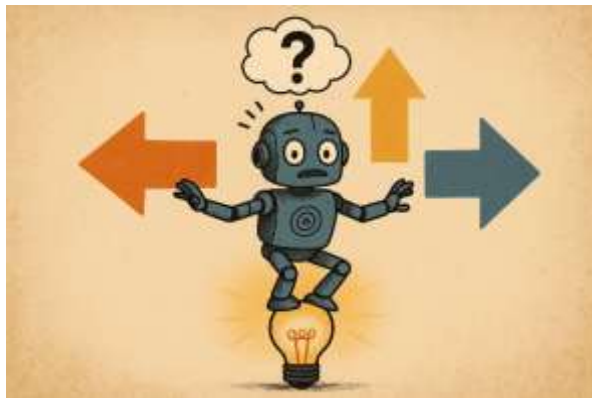


**Continue /
Guided Exercise /
Break?**





**Continue /
Guided Exercise /
Break?**

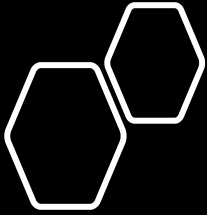


Section 6.4: Guided Exercise - Alerts and Notifications

You should be able to:

- Configure OpenShift to send email notifications.
- Review an alert notification email.
- Silence a firing alert.

Estimated completion time: 10 - 15 minutes



Section 6.5: Lab OpenShift Monitoring

You should be able to:

Configure the OpenShift alert manager to send email notifications for the NodeCPUOvercommit alert rule.

Review the alert email.

Use the OpenShift monitoring dashboards to identify issues with cluster nodes.

Fix the problematic deployment, by reducing its number of pods to zero.